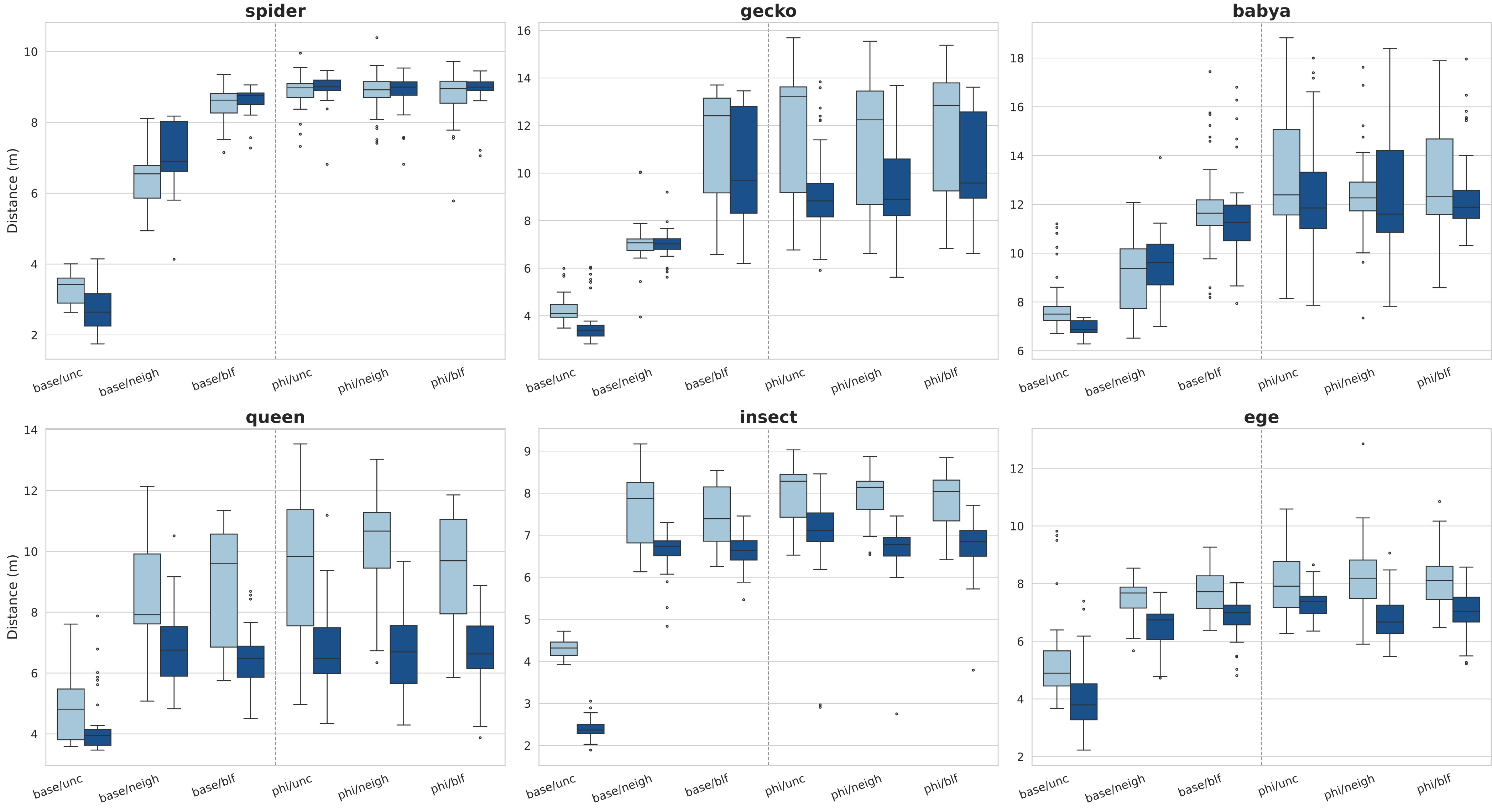


Distance per condition

Always evolved: A (amplitude), X (offset), ψ_{ij} (phase lag per coupled pair). **base:** $\phi_i(0) = 0$ for all oscillators (start synchronized). **phi:** also evolves $\phi_i(0)$. **Fixed:** $\nu = 0.5$ Hz (natural frequency, same for every joint), $w = 1.0$ (coupling weight, same for every coupled pair). **Couplings:** **unc** (no edges), **neigh** (adjacent hinges), **blf** (within-limb). **Fitness function:** **no penalty** $\Rightarrow$ fitness = distance; **with penalty** $\Rightarrow$ fitness = distance $\times$ (1 - dragging). **n = 35** evolved seeds per cell.

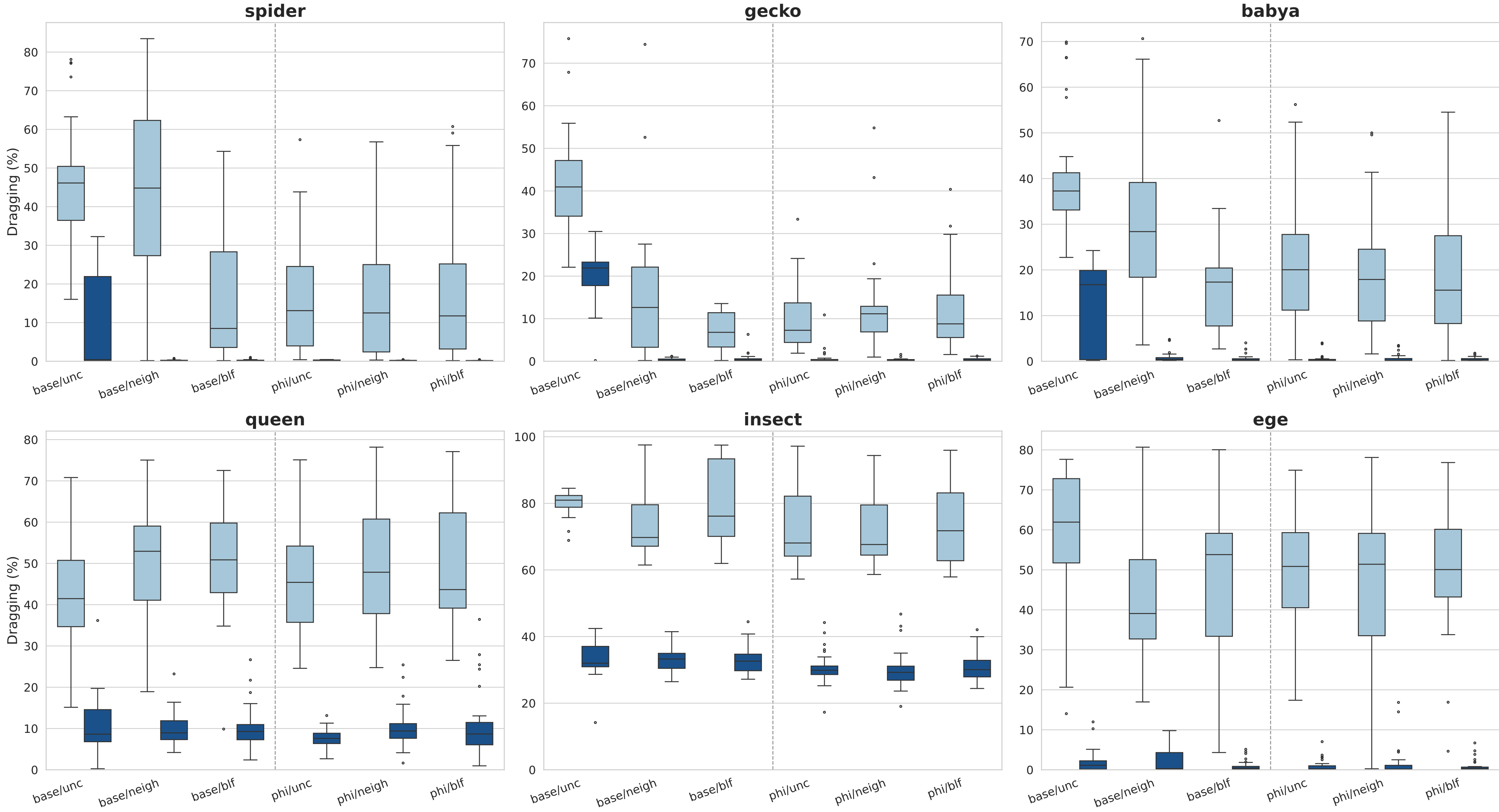
no penalty (distance only) with penalty (distance $\times$ (1 - drag))



Dragging per condition

Always evolved: A (amplitude), X (offset), ψ_i (phase lag per coupled pair). **base:** $\phi_i(0) = 0$ for all oscillators (start synchronized). **phi:** also evolves $\phi_i(0)$. **Fixed:** $\nu = 0.5$ Hz (natural frequency, same for every joint), $w = 1.0$ (coupling weight, same for every coupled pair). **Couplings:** **unc** (no edges), **neigh** (adjacent hinges), **blf** (within-limb). **Fitness function:** **no penalty** $\Rightarrow$ fitness = distance; **with penalty** $\Rightarrow$ fitness = distance $\times$ (1 - dragging). **n = 35** evolved seeds per cell.

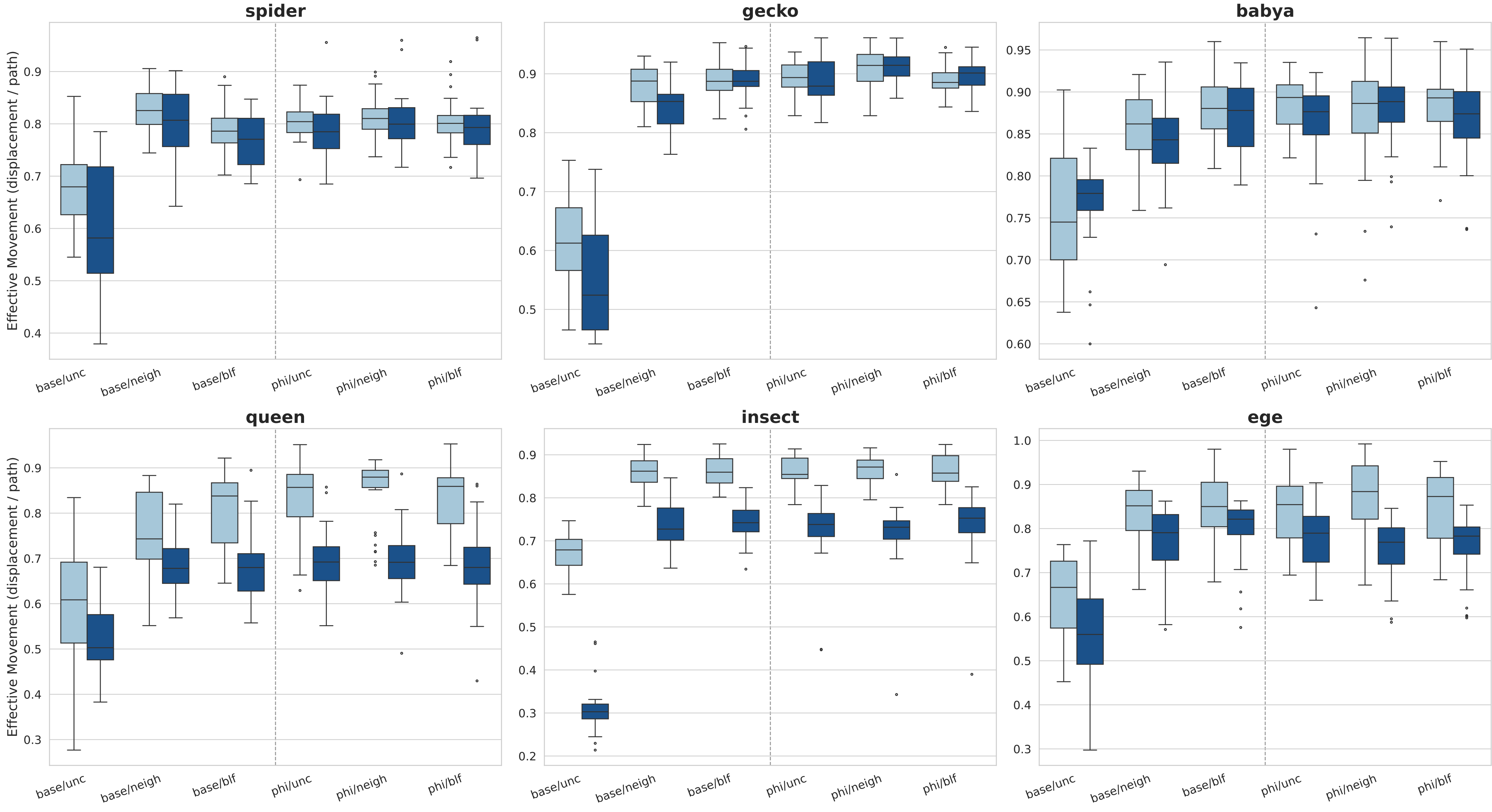
no penalty (distance only) with penalty (distance $\times$ (1 - drag))



Effective Movement per condition

Always evolved: A (amplitude), X (offset), ψ_{ij} (phase lag per coupled pair). **base:** $\phi_i(0) = 0$ for all oscillators (start synchronized). **phi:** also evolves $\phi_i(0)$. **Fixed:** $\nu = 0.5$ Hz (natural frequency, same for every joint), $w = 1.0$ (coupling weight, same for every coupled pair). **Couplings:** **unc** (no edges), **neigh** (adjacent hinges), **blf** (within-limb). **Fitness function:** **no penalty** $\Rightarrow$ fitness = distance; **with penalty** $\Rightarrow$ fitness = distance $\times$ (1 - dragging). **n = 35** evolved seeds per cell.

no penalty (distance only) with penalty (distance $\times$ (1 - drag))



Convergence (best fitness so far across generations)

no penalty (distance only)

Always evolved: A (amplitude), X (offset), ψ_{ij} (phase lag per coupled pair). **base:** $\phi_i(0) = 0$ for all oscillators (start synchronized). **phi:** also evolves $\phi_i(0)$. **Fixed:** $\nu = 0.5$ Hz (natural frequency, same for every joint), $w = 1.0$ (coupling weight, same for every coupled pair).
Line = median across 35 seeds; shaded band = IQR (25-75 percentile). **Fitness function:** no penalty $\Rightarrow$ fitness = distance; with penalty $\Rightarrow$ fitness = distance $\times$ (1 - dragging).

base/unc

base/neigh

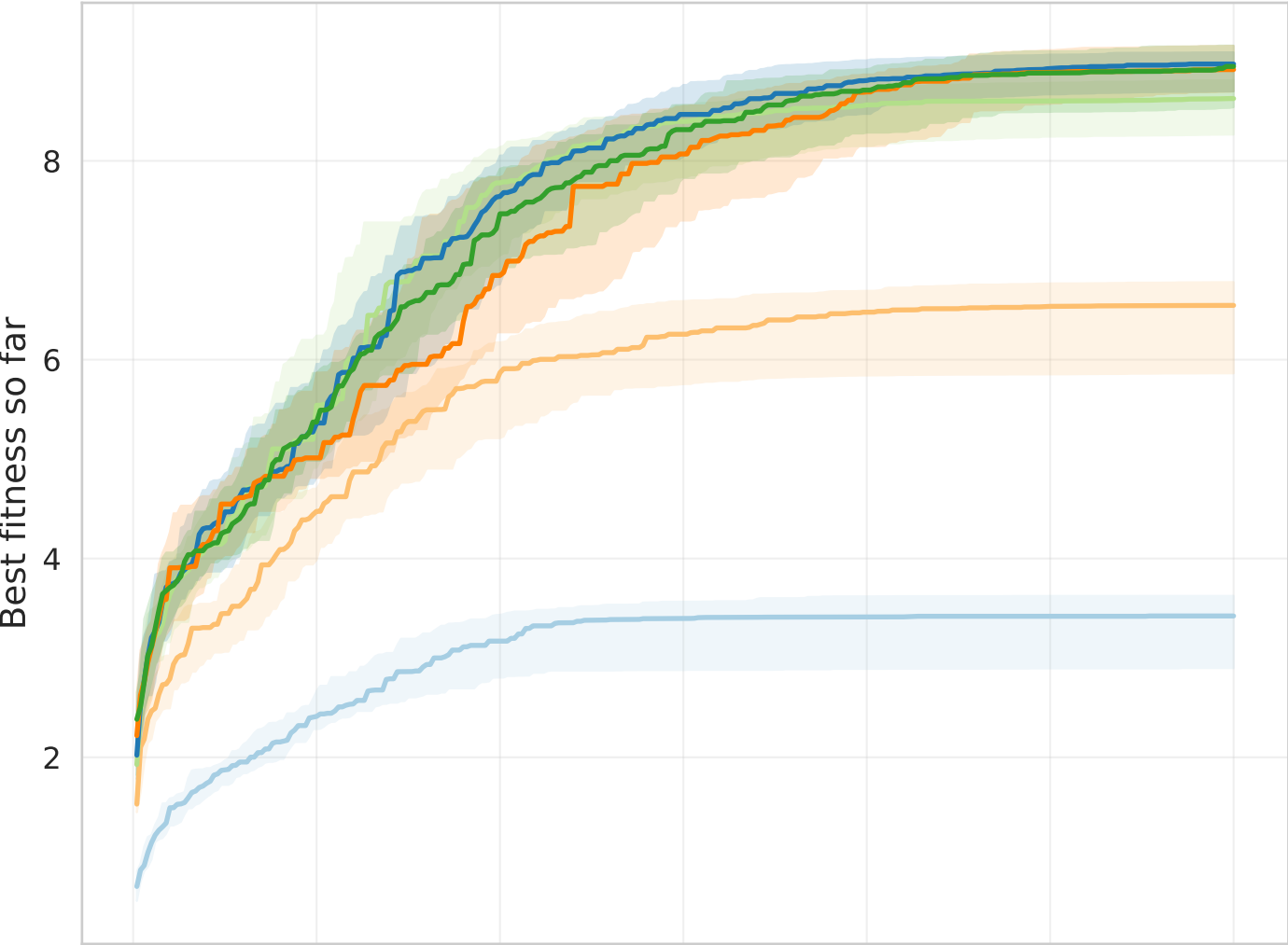
base/blf

phi/unc

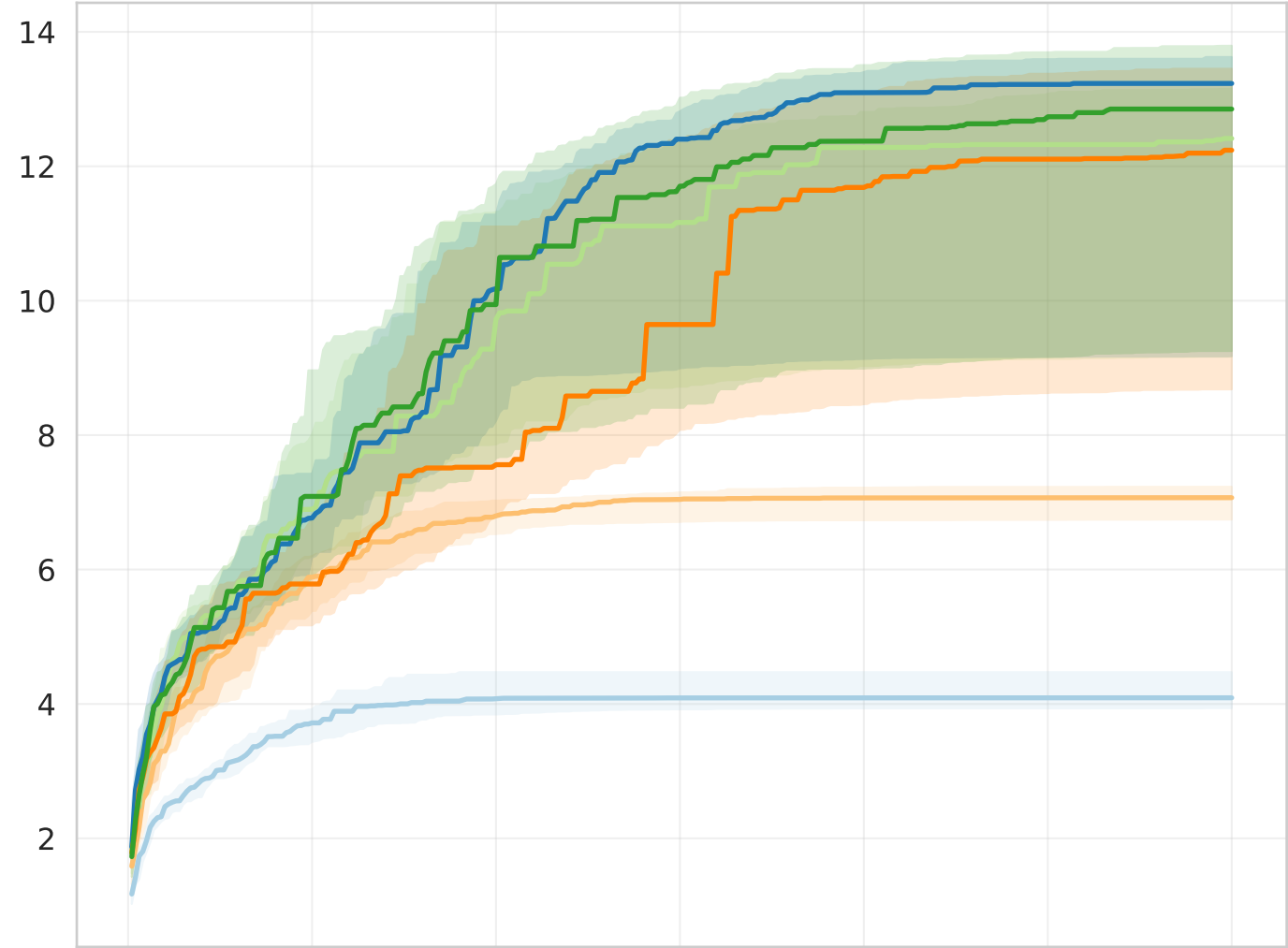
phi/neigh

phi/blf

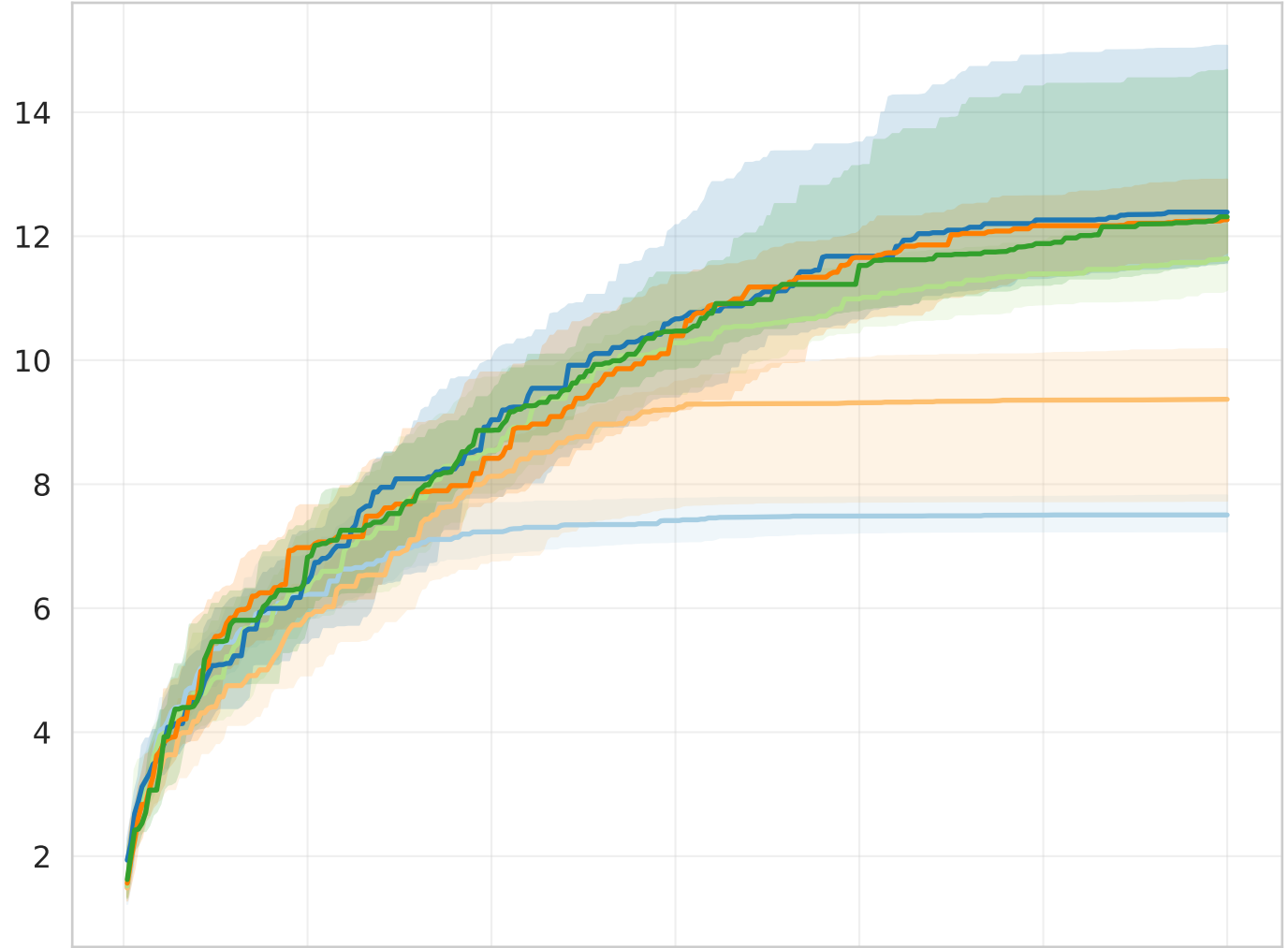
spider



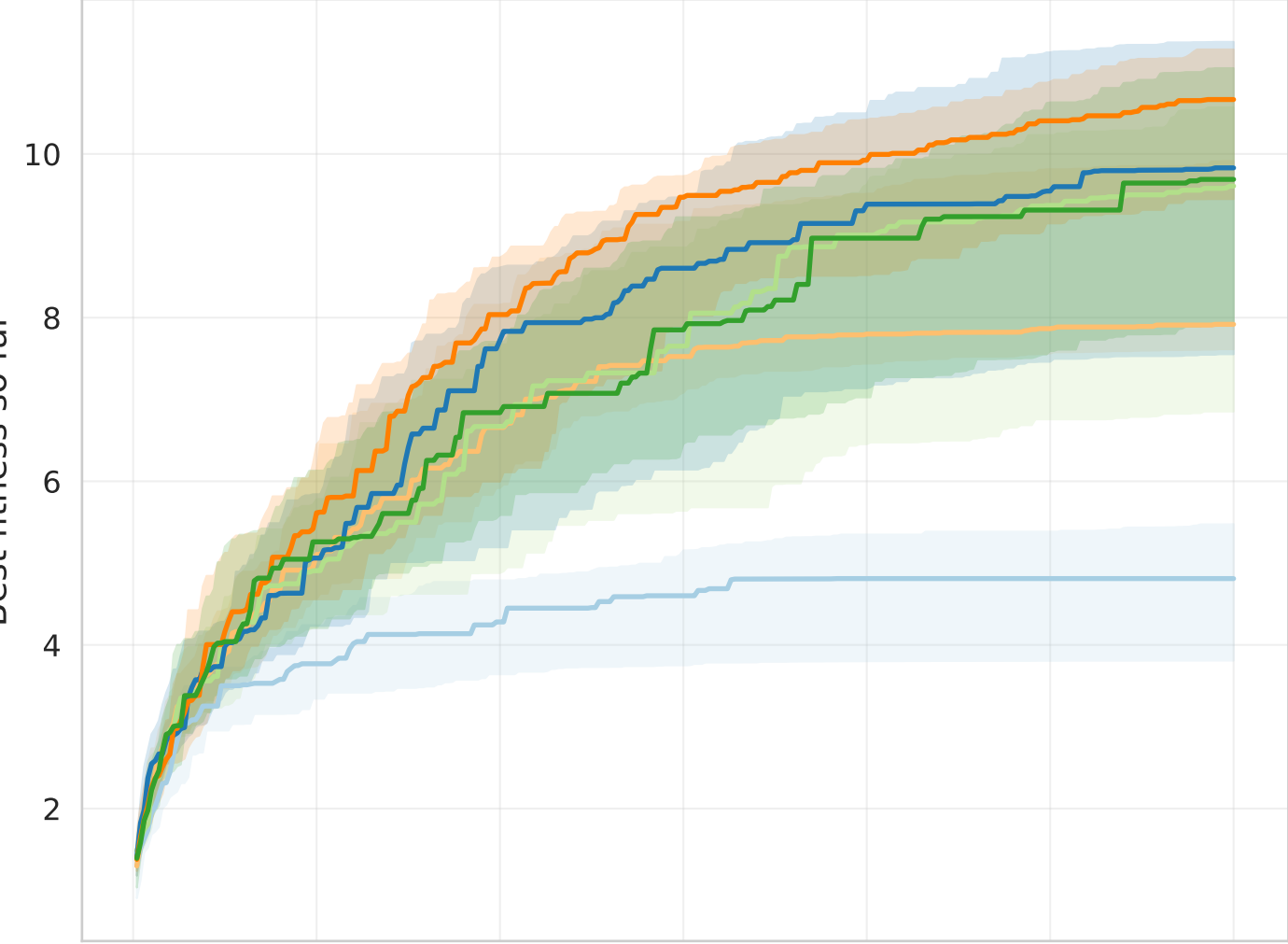
gecko



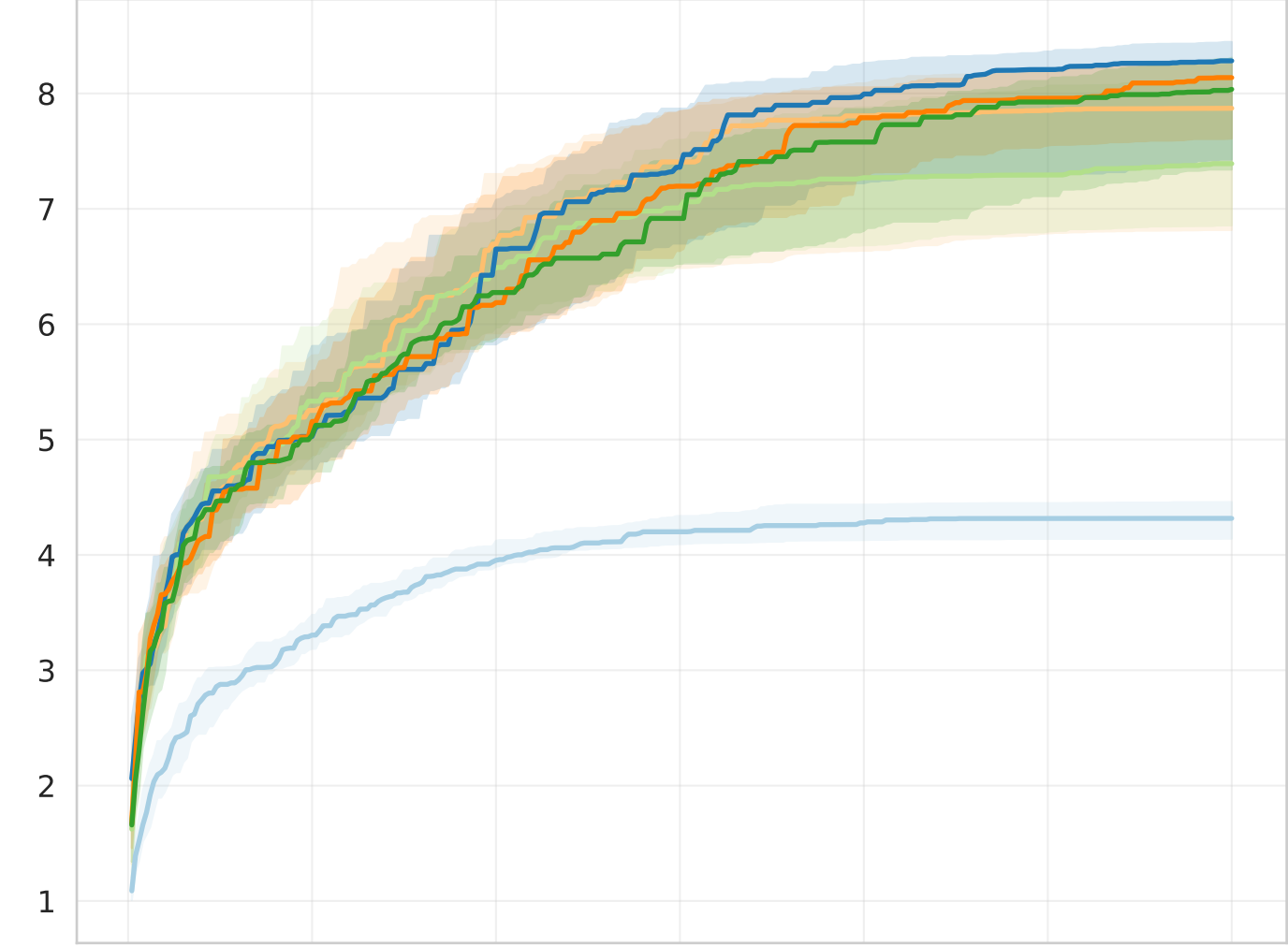
babya



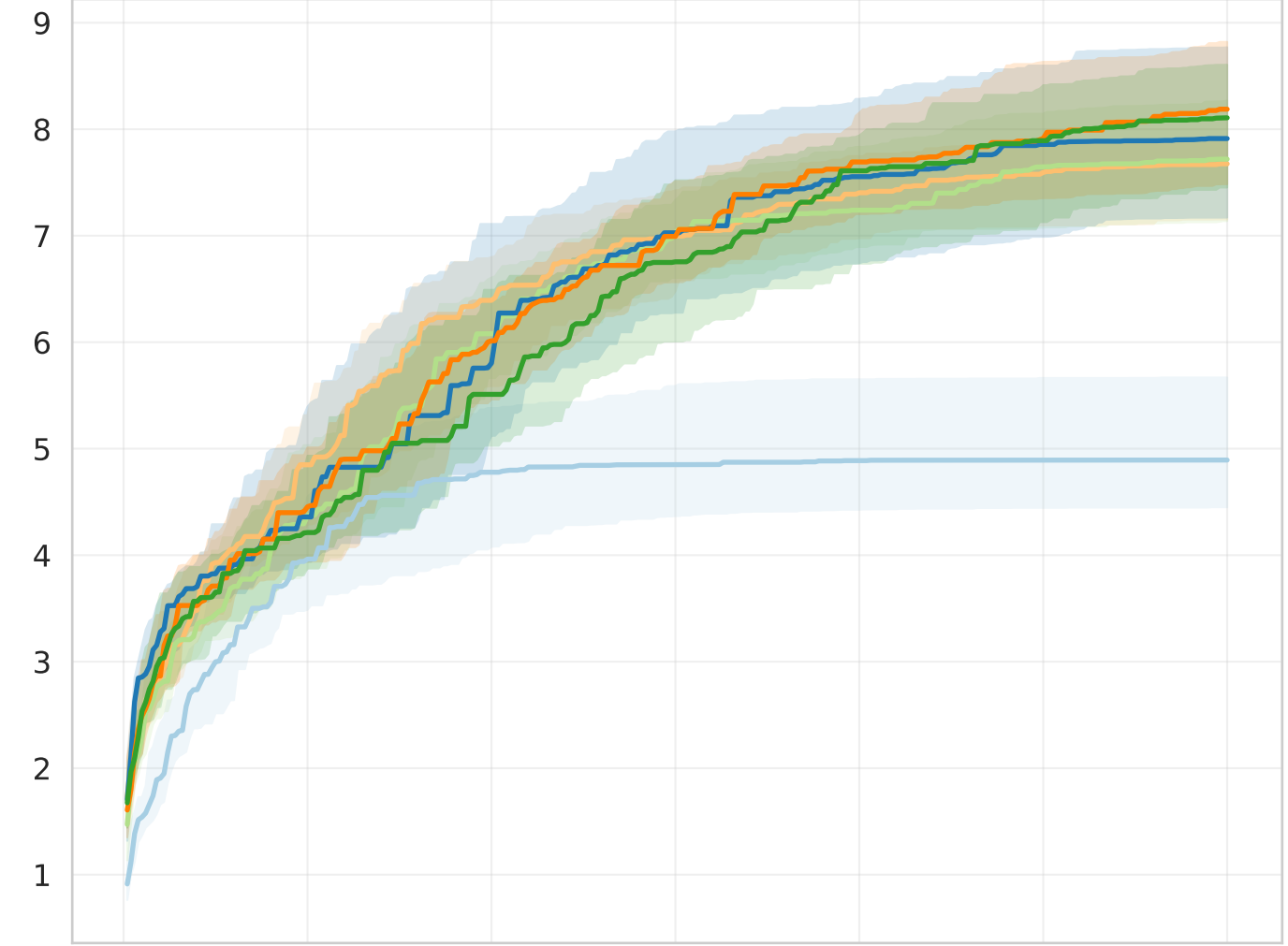
queen



insect



ege



Convergence (best fitness so far across generations)

with penalty (distance × (1 - drag))

Always evolved: A (amplitude), X (offset), ψ_{ij} (phase lag per coupled pair). **base:** $\phi_i(0) = 0$ for all oscillators (start synchronized). **phi:** also evolves $\phi_i(0)$. **Fixed:** $\nu = 0.5$ Hz (natural frequency, same for every joint), $w = 1.0$ (coupling weight, same for every coupled pair).
Line = median across 35 seeds; shaded band = IQR (25-75 percentile). **Fitness function:** no penalty $\Rightarrow$ fitness = distance; with penalty $\Rightarrow$ fitness = distance $\times$ (1 - dragging).

base/unc

base/neigh

base/blf

phi/unc

phi/neigh

phi/blf

